

Session 2:

How Software Works Define software application and its components. Explain software architecture layers: UI ? Business Logic ? Database. Compare desktop, web, and mobile apps. Explain client-server communication with example (browser request ? backend ? database ? response). Trainer Demo: Use Draw.io or Lucidchart AI to visually show data flow (e.g., for “Instagram feed refresh”).

1) Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

1. UI (User Interface) Layer – The part of the application that users interact with (screens, buttons, forms).
2. Business Logic Layer – Processes data, applies rules, and handles the application's core functionality.
3. Database Layer – Stores, retrieves, and manages application data.

2) What is the difference between IPv4 and IPv6? Answer in 2 lines.

- Client: A device or application that requests services or data.
- Server: A system that receives requests and provides data or services.
Example: When you open Instagram on your phone, the Instagram app is the client, and Instagram's computers that store posts and user data are the servers

3) Write 2 differences each between desktop apps, web apps, and mobile apps.

Feature	Desktop App	Web App	Mobile App
Installation	Installed on a computer	Runs in a web browser	Installed on a smartphone/tablet
Internet Requirement	Often works offline	Usually requires internet	Can work online or offline depending on design

Examples:

- Desktop App: Microsoft Word
- Web App: Gmail
- Mobile App: WhatsApp

4) Name any 3 databases and mention whether each is SQL or NoSQL

Database	Type
MySQL	SQL
PostgreSQL	SQL
MongoDB	NoSQL

Client–Server Communication Flow

User (Browser/App)



Request



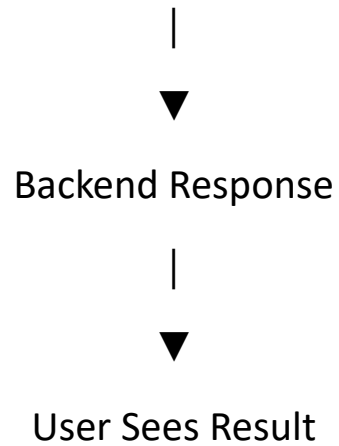
Backend Server



Database



Data Returned



Example (Instagram Feed Refresh):

1. User pulls down to refresh the feed.
2. Instagram app sends a request to the server.
3. Server fetches latest posts from the database.
4. Database returns the data.
5. Server sends the updated feed back to the app.
6. User sees new posts on the screen.